

Inverse Trigonometric Functions

Chapter 2 · All questions with full solutions



Part 1 · Examples

Q

1

Principal value of $\sin^{-1}$



Find the principal value of $\sin^{-1}\left(\frac{1}{\sqrt{2}}\right)$.

Substitute

$$\text{Let } \sin^{-1}\left(\frac{1}{\sqrt{2}}\right) = y.$$

$$\text{Then, } \sin y = \frac{1}{\sqrt{2}}.$$

Key formula

We know that the range of the principal value branch of $\sin^{-1}$ is $\frac{-\pi}{2}$, $\frac{\pi}{2}$ and

$$\sin\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}.$$

Conclusion

Therefore, principal value of $\sin^{-1}\left(\frac{1}{\sqrt{2}}\right)$ is $\frac{\pi}{4}$

Q

2

Principal value of $\cot^{-1}$



Find the principal value of $\cot^{-1}\left(\frac{-1}{\sqrt{3}}\right)$

Substitute

$$\text{Let } \cot^{-1}\left(\frac{-1}{\sqrt{3}}\right) = y.$$

Then,

$$\cot y = \frac{-1}{\sqrt{3}} = -\cot\left(\frac{\pi}{3}\right) = \cot\left(\pi - \frac{\pi}{3}\right) = \cot\left(\frac{2\pi}{3}\right)$$

Key formula

We know that the range of principal value branch of $\cot^{-1}$ is $(0, \pi)$ and $\cot\left(\frac{2\pi}{3}\right)$

$$= \frac{-1}{\sqrt{3}}.$$

Conclusion

Hence, principal value of $\cot^{-1}\left(\frac{-1}{\sqrt{3}}\right)$ is $\frac{2\pi}{3}$

Q

3

Sine inverse double-angle identity



Show that (i) $\sin^{-1}(2x\sqrt{1-x^2}) = 2\sin^{-1}x$, $-\frac{1}{\sqrt{2}} \leq x \leq \frac{1}{\sqrt{2}}$ (ii) $\sin^{-1}(2x\sqrt{1-x^2}) = 2\cos^{-1}x$, $\frac{1}{\sqrt{2}} \leq x \leq 1$

Conclusion

(i) Let $x = \sin\theta$.

Then $\sin^{-1}x = \theta$.

We have

$$\begin{aligned}\sin^{-1}(2x\sqrt{1-x^2}) &= \sin^{-1}(2\sin\theta\sqrt{1-\sin^2\theta}) \\ &= \sin^{-1}(2\sin\theta\cos\theta) = \sin^{-1}(\sin 2\theta) = 2\theta \\ &= 2\sin^{-1}x\end{aligned}$$

(ii) Take $x = \cos\theta$, then proceeding as above, we get, $\sin^{-1}(2x\sqrt{1-x^2}) = 2\cos^{-1}x$

Q

4

Simplify
inverse
tangent
expression



Express $\tan^{-1} \frac{\cos x}{1-\sin x}$, $-\frac{3\pi}{2} < x < \frac{\pi}{2}$ in the simplest form.

Conclusion

We write

$$\begin{aligned}\tan^{-1}\left(\frac{\cos x}{1-\sin x}\right) &= \tan^{-1}\left[\frac{\cos^2 \frac{x}{2} - \sin^2 \frac{x}{2}}{\cos^2 \frac{x}{2} + \sin^2 \frac{x}{2} - 2\sin \frac{x}{2} \cos \frac{x}{2}}\right] \\ &= \tan^{-1}\left[\frac{(\cos \frac{x}{2} + \sin \frac{x}{2})(\cos \frac{x}{2} - \sin \frac{x}{2})}{(\cos \frac{x}{2} - \sin \frac{x}{2})^2}\right] \\ &= \tan^{-1}\left[\frac{\cos \frac{x}{2} + \sin \frac{x}{2}}{\cos \frac{x}{2} - \sin \frac{x}{2}}\right] = \tan^{-1}\left[\frac{1+\tan \frac{x}{2}}{1-\tan \frac{x}{2}}\right] \\ &= \tan^{-1}\left[\tan\left(\frac{\pi}{4} + \frac{x}{2}\right)\right] = \frac{\pi}{4} + \frac{x}{2}\end{aligned}$$

Q

5

Simplify
inverse
cotangent
expression



Write $\cot^{-1}\left(\frac{1}{\sqrt{x^2-1}}\right)$, $x > 1$ in the simplest form.

Conclusion

Let $x = \sec\theta$.

Then $\sqrt{x^2-1} = \sqrt{\sec^2\theta-1} = \tan\theta$.

So, $\cot^{-1} \frac{1}{\sqrt{x^2-1}} = \cot^{-1}(\cot\theta) = \theta = \sec^{-1}x$.

This is the simplest form.

Q

6

Value of
 $\sin^{-1}(\sin x)$
outside
principal
range



Find the value of $\sin^{-1}\left(\sin \frac{3\pi}{5}\right)$

Conclusion

We know that $\sin^{-1}(\sin x) = x$.

So $\sin^{-1}\left(\sin \frac{3\pi}{5}\right) = \frac{3\pi}{5}$.

But $\frac{3\pi}{5} \notin \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

This is outside the principal branch of $\sin^{-1}x$, so it is not yet the final answer.

Now, $\sin\left(\frac{3\pi}{5}\right) = \sin\left(\pi - \frac{3\pi}{5}\right) = \sin \frac{2\pi}{5}$, and $\frac{2\pi}{5} \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

So $\sin^{-1}\left(\sin \frac{3\pi}{5}\right) = \sin^{-1}\left(\sin \frac{2\pi}{5}\right) = \frac{2\pi}{5}$.

Q

1

Principal
value of
 $\sin^{-1}$

(())

Find the principal value of $\sin^{-1}(-\frac{1}{2})$.

Substitute

Let, $\sin^{-1}(-\frac{1}{2}) = y$

Hence,

$$\begin{aligned}\sin y &= (-\frac{1}{2}) \\ &= -\sin(\frac{\pi}{6}) \\ &= \sin(-\frac{\pi}{6})\end{aligned}$$

Key formula

Range of the principal value of $\sin^{-1}(x)$ is $(-\frac{\pi}{2}, \frac{\pi}{2})$

Conclusion

Thus, principal value of $\sin^{-1}(-\frac{1}{2}) = (-\frac{\pi}{6})$.

$-\frac{\pi}{6}$ //

Q

2

Principal
value of
 $\cos^{-1}$

(())

Find the principal value of $\cos^{-1}(\frac{\sqrt{3}}{2})$.

Substitute

Let,

$\cos^{-1}(\frac{\sqrt{3}}{2}) = y$

Hence,

$$\begin{aligned}\cos y &= (\frac{\sqrt{3}}{2}) \\ &= \cos \frac{\pi}{6}\end{aligned}$$

Key formula

Range of the principal value of $\cos^{-1}(x)$ is $(0, \pi)$.

Conclusion

Thus, principal value of $\cos^{-1}(\frac{\sqrt{3}}{2}) = (\frac{\pi}{6})$

$\frac{\pi}{6}$ //

Q**3**Principal
value of
 $\operatorname{cosec}^{-1}$ Find the principal value of $\operatorname{cosec}^{-1}(2)$.**Substitute**

Let, $\operatorname{cosec}^{-1}(2) = y$

Hence,

$$\begin{aligned}\operatorname{cosec} y &= 2 \\ &= \operatorname{cosec}\left(\frac{\pi}{6}\right)\end{aligned}$$

Key formula

Range of the principal value of $\operatorname{cosec}^{-1}(x) = \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] - \{0\}$

**Conclusion**

Thus, principal value of $\operatorname{cosec}^{-1}(2) = \left(\frac{\pi}{6}\right)$.

$$\left(\frac{\pi}{6}\right) //$$

Q**4**Principal
value of
 $\tan^{-1}$ Find the principal value of $\tan^{-1}(-\sqrt{3})$ **Substitute**

Let, $\tan^{-1}(-\sqrt{3}) = y$

Hence,

$$\begin{aligned}\tan y &= -\sqrt{3} \\ &= -\tan\left(\frac{\pi}{3}\right) \\ &= \tan\left(-\frac{\pi}{3}\right)\end{aligned}$$

Key formula

Range of the principal value of $\tan^{-1}(x) = \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

**Conclusion**

Thus, principal value of $\tan^{-1}(-\sqrt{3}) = \left(-\frac{\pi}{3}\right)$.

$$\left(-\frac{\pi}{3}\right) //$$

Q**5**Principal
value of
 $\cos^{-1}$ **(())**Find the principal value of $\cos^{-1}\left(-\frac{1}{2}\right)$ **Substitute**

Let, $\cos^{-1}\left(-\frac{1}{2}\right) = y$

Hence,

$$\begin{aligned}
 \cos y &= -\frac{1}{2} \\
 &= -\cos\left(\frac{\pi}{3}\right) \\
 &= \cos\left(\pi - \frac{\pi}{3}\right) \\
 &= \cos\left(\frac{2\pi}{3}\right)
 \end{aligned}$$

Key formula

Range of the principal value of $\cos^{-1}(x) = [0, \pi]$

Conclusion

Thus, principal value of $\cos^{-1}\left(-\frac{1}{2}\right) = \left(\frac{2\pi}{3}\right)$.

$$\frac{2\pi}{3} //$$

Q

6

Principal
value of
 $\tan^{-1}$



Find the principal value of $\tan^{-1}(-1)$

Substitute

Let, $\tan^{-1}(-1) = y$

Hence,

$$\begin{aligned}
 \tan y &= -1 \\
 &= -\tan\left(\frac{\pi}{4}\right) \\
 &= \tan\left(-\frac{\pi}{4}\right)
 \end{aligned}$$

Key formula

Range of the principal value of $\tan^{-1}(x) = \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

Conclusion

Thus, principal value of $\tan^{-1}(-1) = \left(-\frac{\pi}{4}\right)$.

$$-\frac{\pi}{4} //$$

Q

7

Principal
value of
 $\sec^{-1}$



Find the principal value of $\sec^{-1}\left(\frac{2}{\sqrt{3}}\right)$

Substitute

Let, $\sec^{-1}\left(\frac{2}{\sqrt{3}}\right) = y$

Hence,

$$\begin{aligned}
 \sec y &= \frac{2}{\sqrt{3}} \\
 &= \sec\left(\frac{\pi}{6}\right)
 \end{aligned}$$

Key formula

Range of the principal value of $\sec^{-1}(x) = [0, \pi] - \{ \frac{\pi}{2} \}$

**Conclusion**

Thus, principal value of $\sec^{-1}(\frac{2}{\sqrt{3}}) = (\frac{\pi}{6})$.

$$\frac{\pi}{6} //$$

Q**8**

Principal
value of
 $\cot^{-1}$



Find the principal value of $\cot^{-1}(\sqrt{3})$

Substitute

Let, $\cot^{-1}(\sqrt{3}) = y$

Hence,

$$\begin{aligned} \cot y &= \sqrt{3} \\ &= \cot\left(\frac{\pi}{6}\right) \end{aligned}$$

Key formula

Range of the principal value of $\cot^{-1}(x) = (0, \pi)$

**Conclusion**

Thus, principal value of $\cot^{-1}(\sqrt{3}) = (\frac{\pi}{6})$.

$$\frac{\pi}{6} //$$

Q**9**

Principal
value of
 $\cos^{-1}$

(())

Find the principal value of $\cos^{-1}(-\frac{1}{\sqrt{2}})$

Substitute

Let, $\cos^{-1}(-\frac{1}{\sqrt{2}}) = y$

Hence,

$$\begin{aligned} \cos y &= -\frac{1}{\sqrt{2}} \\ &= -\cos\left(\frac{\pi}{4}\right) \\ &= \cos\left(-\frac{\pi}{4}\right) \\ &= \cos\left(\pi - \frac{\pi}{4}\right) \\ &= \cos\left(\frac{3\pi}{4}\right) \end{aligned}$$

Key formula

Range of the principal value of $\cos^{-1}(x) = [0, \pi]$



Conclusion

Thus, principal value of $\cos^{-1}(-\frac{1}{\sqrt{2}}) = (\frac{3\pi}{4})$.

$$\frac{3\pi}{4} //$$

Q**10**

Principal
value of
 $\operatorname{cosec}^{-1}$



Find the principal value of $\operatorname{cosec}^{-1}(-\sqrt{2})$

Substitute

Let, $\operatorname{cosec}^{-1}(-\sqrt{2}) = y$

Hence,

$$\begin{aligned}\operatorname{cosec} y &= -\sqrt{2} \\ &= -\operatorname{cosec}\left(\frac{\pi}{4}\right) \\ &= \operatorname{cosec}\left(-\frac{\pi}{4}\right)\end{aligned}$$

Key formula

Range of the principal value of $\operatorname{cosec}^{-1}(x) = [-\frac{\pi}{2}, \frac{\pi}{2}] - \{0\}$

**Conclusion**

Thus, principal value of $\operatorname{cosec}^{-1}(-\sqrt{2}) = (-\frac{\pi}{4})$.

$$-\frac{\pi}{4} //$$

Q**11**

Sum of
inverse trig
values



Find the value of $\tan^{-1}(1) + \cos^{-1}(-\frac{1}{2}) + \sin^{-1}(\frac{1}{2})$

Given

Let, $\tan^{-1}(1) = x$

Hence,

$$\begin{aligned}\tan x &= 1 \\ &= \tan\left(\frac{\pi}{4}\right)\end{aligned}$$

Therefore,

$$\tan^{-1}(1) = \left(\frac{\pi}{4}\right)$$

Substitute

Now, let $\cos^{-1}(-\frac{1}{2}) = y$

Hence,

$$\begin{aligned}\cos y &= -\frac{1}{2} \\ &= -\cos\left(\frac{\pi}{3}\right) \\ &= \cos\left(\pi - \frac{\pi}{3}\right) \\ &= \cos\left(\frac{2\pi}{3}\right)\end{aligned}$$

Therefore,

$$\cos^{-1}\left(-\frac{1}{2}\right) = \frac{2\pi}{3}$$

$$\text{Again, let } \sin^{-1}\left(-\frac{1}{2}\right) = z$$

Hence,

$$\begin{aligned}\sin z &= -\frac{1}{2} \\ &= -\sin\left(\frac{\pi}{6}\right) \\ &= \sin\left(-\frac{\pi}{6}\right)\end{aligned}$$

Therefore,

$$\sin^{-1}\left(-\frac{1}{2}\right) = -\frac{\pi}{6}$$

Conclusion

Thus,

$$\begin{aligned}\tan^{-1}(1) + \cos^{-1}\left(-\frac{1}{2}\right) + \sin^{-1}\left(-\frac{1}{2}\right) &= \frac{\pi}{4} + \frac{2\pi}{3} - \frac{\pi}{6} \\ &= \frac{3\pi + 8\pi - 2\pi}{12} \\ &= \frac{9\pi}{12} \\ &= \frac{3\pi}{4}\end{aligned}$$

$\frac{3\pi}{4} //$

Q

12

Sum of
inverse trig
values



Find the value of $\cos^{-1} \frac{1}{2} + 2 \sin^{-1} \frac{1}{2}$

Given

$$\text{Let, } \tan^{-1}(1) = x$$

Hence,

$$\begin{aligned}\cos x &= \frac{1}{2} \\ &= \cos\left(\frac{\pi}{3}\right)\end{aligned}$$

Therefore,

$$\cos^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{3}$$

Substitute

$$\text{Let, } \sin^{-1}\left(\frac{1}{2}\right) = y$$

Hence,

$$\begin{aligned}\sin y &= \frac{1}{2} \\ &= \sin\left(\frac{\pi}{6}\right)\end{aligned}$$

Therefore,

$$\sin^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{6}$$

Conclusion

Thus

$$\begin{aligned} \cos^{-1}\left(\frac{1}{2}\right) + 2\sin^{-1}\left(\frac{1}{2}\right) &= \frac{\pi}{3} + 2\left(\frac{\pi}{6}\right) \\ &= \frac{2\pi}{3} \end{aligned}$$

$$\frac{2\pi}{3} //$$

Q

13

Range of $\sin^{-1}$
(multiple choice)

(())

If $\sin^{-1} x = y$, then (A) $0 \leq y \leq \pi$ (B) $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$ (C) $0 < y < \pi$ (D) $-\frac{\pi}{2} < y < \frac{\pi}{2}$

Substitute

It is given that $\sin^{-1} x = y$

Key formula

Range of the principal value of $\sin^{-1} x = \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

Conclusion

Thus, $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$

The answer is B.

Option (B) //

Q

14

Difference of
inverse trig
values
(multiple choice)

(())

$\tan^{-1} \sqrt{3} - \sec^{-1}(-2)$ is equal to (A) π (B) $-\frac{\pi}{3}$ (C) $\frac{\pi}{3}$ (D) $\frac{2\pi}{3}$

Given

Let $\tan^{-1}(\sqrt{3}) = x$

Hence,

$$\begin{aligned} \tan x &= \sqrt{3} \\ &= \tan\left(\frac{\pi}{3}\right) \end{aligned}$$

Key formula

Range of the principal value of $\tan^{-1} x = \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

Substitute

Therefore, $\tan^{-1}(\sqrt{3}) = \left(\frac{\pi}{3}\right)$

Let $\sec^{-1}(-2) = y$

Hence,

$$\begin{aligned}
 \sec y &= (-2) \\
 &= -\sec\left(\frac{\pi}{3}\right) \\
 &= \sec\left(-\frac{\pi}{3}\right) \\
 &= \sec\left(\pi - \frac{\pi}{3}\right) \\
 &= \sec\left(\frac{2\pi}{3}\right)
 \end{aligned}$$

Key formula

Range of the principal value of $\sec^{-1} x = [0, \pi] - \left\{ \frac{\pi}{2} \right\}$



Conclusion

Therefore, $\sec^{-1}(-2) = \frac{2\pi}{3}$

Thus,

$$\begin{aligned}
 \tan^{-1} \sqrt{3} - \sec^{-1}(-2) &= \frac{\pi}{3} - \frac{2\pi}{3} \\
 &= -\frac{\pi}{3}
 \end{aligned}$$

Option (B) //

Q

1

Prove triple-angle sine identity



Prove $3 \sin^{-1} x = \sin^{-1}(3x - 4x^3)$, $x \in [-\frac{1}{2}, \frac{1}{2}]$.

Conclusion

Let $x = \sin \theta$ in

Hence, $\sin^{-1}(x) =$

Now,

$$\begin{aligned} \text{RHS} &= \sin^{-1}(3x - 4x^3) \\ &= \sin^{-1}(3\sin\theta - 4\sin^3\theta) \\ &= \sin^{-1}(\sin 3\theta) \\ &= \theta \\ &= 3 \sin^{-1} x \\ &= \text{LHS} \end{aligned}$$

Q

2

Prove triple-angle cosine identity



Prove $3 \cos^{-1} x = \cos^{-1}(4x^3 - 3x)$, $x \in [\frac{1}{2}, 1]$.

Conclusion

Let $x = \cos \theta$ in

Hence, $\cos^{-1}(x) =$

Now,

$$\begin{aligned} \text{RHS} &= \cos^{-1}(4x^3 - 3x) \\ &= \cos^{-1}(4\cos^3\theta - 3\cos\theta) \\ &= \cos^{-1}(\cos 3\theta) \\ &= \theta \\ &= 3 \cos^{-1} x \\ &= \text{LHS} \end{aligned}$$

Q

3

Simplify inverse tangent expression

(())

$\tan^{-1} \frac{\sqrt{1+x^2}-1}{x}$, $x \neq 0$

Conclusion

Let $x = \tan \theta \Rightarrow \theta = \tan^{-1} x$

Hence,

$$\begin{aligned}
 \tan^{-1} \frac{\sqrt{(1+x^2)}-1}{x} &= \tan^{-1} \left(\frac{\sqrt{(1+\tan^2 \theta)}-1}{\tan \theta} \right) \\
 &= \tan^{-1} \left(\frac{\sec \theta - 1}{\tan \theta} \right) \\
 &= \tan^{-1} \left(\frac{1 - \cos \theta}{\sin \theta} \right) \\
 &= \tan^{-1} \left(\frac{2 \sin^2 \frac{\theta}{2}}{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}} \right) \\
 &= \tan^{-1} \left(\tan \frac{\theta}{2} \right) \\
 &= \frac{\theta}{2} \\
 &= \frac{1}{2} \tan^{-1} x \\
 &= \frac{1}{2} \tan^{-1} x \quad ///
 \end{aligned}$$

Q

4

Simplify
inverse
tangent
expression



$$\tan^{-1} \left(\sqrt{\frac{1 - \cos x}{1 + \cos x}} \right), 0 < x < \pi$$

Conclusion

$$\text{Since, } 1 - \cos x = 2 \sin^2 \frac{x}{2} \text{ and } 1 + \cos x = 2 \cos^2 \frac{x}{2}$$

Hence,

$$\begin{aligned}
 \tan^{-1} \left(\sqrt{\frac{1 - \cos x}{1 + \cos x}} \right) &= \tan^{-1} \left(\sqrt{\frac{2 \sin^2 \frac{x}{2}}{2 \cos^2 \frac{x}{2}}} \right) \\
 &= \tan^{-1} \left(\frac{\sin \frac{x}{2}}{\cos \frac{x}{2}} \right) \\
 &= \tan^{-1} \left(\tan \frac{x}{2} \right) \\
 &= \frac{x}{2} \\
 &= \frac{x}{2} \quad ///
 \end{aligned}$$

Q

5

Simplify
inverse
tangent
expression



$$\tan^{-1} \left(\frac{\cos x - \sin x}{\cos x + \sin x} \right), \frac{-\pi}{4} < x < \frac{3\pi}{4}$$

Conclusion

$$\begin{aligned}
 \tan^{-1}\left(\frac{\cos x - \sin x}{\cos x + \sin x}\right) &= \tan^{-1}\left(\frac{\frac{\cos x - \sin x}{\cos x}}{\frac{\cos x + \sin x}{\cos x}}\right) \\
 &= \tan^{-1}\left(\frac{1 - \frac{\sin x}{\cos x}}{1 + \frac{\sin x}{\cos x}}\right) \\
 &= \tan^{-1}\left(\frac{1 - \tan x}{1 + \tan x}\right) \\
 &= \tan^{-1}(1) - \tan^{-1}(\tan x) \\
 &= \frac{\pi}{4} - x \\
 &\boxed{\frac{\pi}{4} - x} \quad ///
 \end{aligned}$$

Q

6

Simplify
inverse
tangent
expression



$$\tan^{-1} \frac{x}{\sqrt{(a^2 - x^2)}}, |x| < a$$

Conclusion

Let

$$x = a \sin \theta \Rightarrow \theta = \sin^{-1}\left(\frac{x}{a}\right)$$

Hence,

$$\begin{aligned}
 \tan^{-1} \frac{x}{\sqrt{(a^2 - x^2)}} &= \tan^{-1}\left(\frac{a \sin \theta}{\sqrt{(a^2 - a^2 \sin^2 \theta)}}\right) \\
 &= \tan^{-1}\left(\frac{a \sin \theta}{a \sqrt{(1 - \sin^2 \theta)}}\right) \\
 &= \tan^{-1}\left(\frac{a \sin \theta}{a \cos \theta}\right) \\
 &= \tan^{-1}(\tan \theta) \\
 &= \theta \\
 &= \sin^{-1} \frac{x}{a}
 \end{aligned}$$

$$\boxed{\sin^{-1} \frac{x}{a}} \quad ///$$

Q

7

Simplify
inverse
tangent
expression

((

$$\tan^{-1}\left(\frac{3a^2x - x^3}{a^3 - 3ax^2}\right), a > 0; \frac{-a}{\sqrt{3}} < x < \frac{a}{\sqrt{3}}$$

Conclusion

$$\text{Let } x = a \tan \theta \Rightarrow \theta = \tan^{-1}\left(\frac{x}{a}\right)$$

Hence,

$$\begin{aligned}
 \tan^{-1}\left(\frac{3a^2x-x^3}{a^3-3ax^2}\right) &= \tan^{-1}\left(\frac{3a^2 \cdot a \tan\theta - a^3 \tan^3\theta}{a^3 - 3a \cdot a^2 \tan^2\theta}\right) \\
 &= \tan^{-1}\left(\frac{3a^3 \tan\theta - a^3 \tan^3\theta}{a^3 - 3a^3 \tan^2\theta}\right) \\
 &= \tan^{-1}(\tan 3\theta) \\
 &= 3\theta \\
 &= 3 \tan^{-1} \frac{x}{a} \\
 &\boxed{3 \tan^{-1} \frac{x}{a}} ///
 \end{aligned}$$

Q

8

Value of
nested
inverse trig
expression



$$\tan^{-1}\left[2 \cos\left(2 \sin^{-1} \frac{1}{2}\right)\right]$$

Conclusion

$$\text{Let } \sin^{-1} \frac{1}{2} = x$$

Hence,

$$\begin{aligned}
 \sin x &= \frac{1}{2} \\
 &= \sin\left(\frac{\pi}{6}\right) \\
 x &= \frac{\pi}{6} \\
 \sin^{-1}\left(\frac{1}{2}\right) &= \frac{\pi}{6}
 \end{aligned}$$

Therefore,

$$\begin{aligned}
 \tan^{-1}\left[2 \cos\left(2 \sin^{-1} \frac{1}{2}\right)\right] &= \tan^{-1}\left[2 \cos\left(2 \times \frac{\pi}{6}\right)\right] \\
 &= \tan^{-1}\left[2 \cos \frac{\pi}{3}\right] \\
 &= \tan^{-1}\left[2 \times \frac{1}{2}\right] \\
 &= \tan^{-1}[1] \\
 &= \frac{\pi}{4}
 \end{aligned}$$

$$\boxed{\frac{\pi}{4}} ///$$

Q

9

Value of
combined
inverse trig
expression



$$\tan \frac{1}{2} \left[\sin^{-1} \frac{2x}{1+x^2} + \cos^{-1} \frac{1-y^2}{1+y^2} \right], |x| < 1, y > 0 \text{ and } x y < 1$$

Given

$$\text{Let } x = \tan\theta \Rightarrow \theta = \tan^{-1} x$$

Hence,

$$\begin{aligned}
 \sin^{-1} \frac{2x}{1+x^2} &= \sin^{-1}\left(\frac{2 \tan\theta}{1+\tan^2\theta}\right) \\
 &= \sin^{-1}(\sin 2\theta) \\
 &= 2\theta \\
 &= 2 \tan^{-1} x
 \end{aligned}$$

Substitute

Now, let $y = \tan \varphi \Rightarrow \varphi = \tan^{-1} y$

Hence,

$$\begin{aligned}\cos^{-1} \frac{1-y^2}{1+y^2} &= \cos^{-1} \left(\frac{1-\tan^2 \varphi}{1+\tan^2 \varphi} \right) \\ &= \cos^{-1}(\cos 2\varphi) \\ &= 2\varphi \\ &= 2 \tan^{-1} y\end{aligned}$$

Conclusion

Therefore,

$$\begin{aligned}\tan \frac{1}{2} (\sin^{-1} \frac{2x}{1+x^2} + \cos^{-1} \frac{1-y^2}{1+y^2}) &= \tan \frac{1}{2} (2 \tan^{-1} x + 2 \tan^{-1} y) \\ &= \tan(\tan^{-1} x + \tan^{-1} y) \\ &= \tan\left[\tan^{-1}\left(\frac{x+y}{1-xy}\right)\right] \\ &= \left(\frac{x+y}{1-xy}\right)\end{aligned}$$

$\frac{x+y}{1-xy}$

Q**10**

Value of
 $\sin^{-1}(\sin x)$
outside
principal
range



$$\sin^{-1}\left(\sin \frac{2\pi}{3}\right)$$

Conclusion

Since, $\sin(\theta)$

Therefore,

$$\begin{aligned}\sin^{-1}\left(\sin \frac{2\pi}{3}\right) &= \sin^{-1}\left[\sin\left(\pi - \frac{2\pi}{3}\right)\right] \\ &= \sin^{-1}\left(\sin \frac{\pi}{3}\right) \\ &= \frac{\pi}{3}\end{aligned}$$

$\frac{\pi}{3}$

Q**11**

Value of
 $\tan^{-1}(\tan x)$
outside
principal
range

((

$$\tan^{-1}\left(\tan \frac{3\pi}{4}\right)$$

Conclusion

Since, $\tan(\theta)$

Therefore,

$$\begin{aligned}\tan^{-1}\left(\tan \frac{3\pi}{4}\right) &= \tan^{-1}\left[-\tan\left(\pi - \frac{3\pi}{4}\right)\right] \\ &= \tan^{-1}\left[-\tan\left(\pi - \frac{\pi}{4}\right)\right] \\ &= \tan^{-1}\left[-\tan\left(\frac{\pi}{4}\right)\right] \\ &= \tan^{-1}\left[\tan\left(-\frac{\pi}{4}\right)\right] \\ &= \left(-\frac{\pi}{4}\right)\end{aligned}$$

$$-\frac{\pi}{4} //$$

Q

12

Value of
combined
inverse trig
expression



$$\tan\left(\sin^{-1} \frac{3}{5} + \cot^{-1} \frac{3}{2}\right)$$

Given

$$\text{Let } \sin^{-1} \frac{3}{5} = x \Rightarrow \sin x = \frac{3}{5}$$

Then,

$$\Rightarrow \cos x = \sqrt{1 - \sin^2 x} = \frac{4}{5}$$

$$\Rightarrow \sec x = \frac{5}{4}$$

Therefore,

$$\tan x = \sqrt{\sec^2 x - 1}$$

$$= \sqrt{\left(\frac{25}{16} - 1\right)}$$

$$= \frac{3}{4}$$

$$x = \tan^{-1} \frac{3}{4}$$

$$\sin^{-1} \frac{3}{5} = \tan^{-1} \frac{3}{4}$$

Substitute

Now,

$$\cot^{-1} \frac{3}{2} = \tan^{-1} \frac{2}{3}$$

Conclusion

Thus, by using (1) and (2)

$$\tan\left(\sin^{-1} \frac{3}{5} + \cot^{-1} \frac{3}{2}\right) = \tan\left(\tan^{-1} \frac{3}{4} + \tan^{-1} \frac{2}{3}\right)$$

$$= \tan\left[\tan^{-1} \frac{\frac{3}{4} + \frac{2}{3}}{1 - \frac{3}{4} \cdot \frac{2}{3}}\right]$$

$$= \tan\left(\tan^{-1} \frac{17}{6}\right)$$

$$= \frac{17}{6}$$

$$\frac{17}{6} //$$

Q**13**

Value of $\cos^{-1}(\cos x)$ outside principal range (multiple choice)



$\cos^{-1}(\cos \frac{7\pi}{6})$ is equal to (A) $\frac{7\pi}{6}$ (B) $\frac{5\pi}{6}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{6}$

Conclusion

$$\begin{aligned}\cos^{-1}(\cos \frac{7\pi}{6}) &= \cos^{-1}(\cos \frac{-7\pi}{6}) \\ &= \cos^{-1}[\cos(2\pi - \frac{7\pi}{6})] \\ &= \cos^{-1}[\cos(\frac{5\pi}{6})] \\ &= \frac{5\pi}{6}\end{aligned}$$

Thus, the correct option is B.

Option (B) ///

Q**14**

Value of trig expression with inverse sine (multiple choice)



$\sin(\frac{\pi}{3} - \sin^{-1}(-\frac{1}{2}))$ is equal to (A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) $\frac{1}{4}$ (D) 1

Conclusion

$$\text{Let } \sin^{-1}(-\frac{1}{2}) = x$$

Hence,

$$\begin{aligned}\sin x &= -\frac{1}{2} \\ &= -\sin \frac{\pi}{6} \\ &= \sin(-\frac{\pi}{6}) \\ x &= -\frac{\pi}{6}\end{aligned}$$

Since, Range of principal value of $\sin^{-1}(x) = [-\frac{\pi}{2}, \frac{\pi}{2}]$.

Therefore,

$$\sin^{-1}(-\frac{1}{2}) = -\frac{\pi}{6}$$

Then,

$$\begin{aligned}\sin(\frac{\pi}{3} - \sin^{-1}(-\frac{1}{2})) &= \sin(\frac{\pi}{3} + \frac{\pi}{6}) \\ &= \sin(\frac{\pi}{2}) \\ &= 1\end{aligned}$$

Thus, the correct option is D.

Option (D) ///

Q**15**

**Difference of
inverse trig
values
(multiple
choice)**

()

$\tan^{-1} \sqrt{3} - \cot^{-1}(-\sqrt{3})$ is equal to (A) π (B) $-\frac{\pi}{2}$ (C) 0 (D) $2\sqrt{3}$

Conclusion

Let $\tan^{-1} \sqrt{3} = x$

Hence,

$$\tan x = \sqrt{3} = \tan \frac{\pi}{3}, \text{ where } \frac{\pi}{3} \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

$$\text{Therefore, } \tan^{-1} \sqrt{3} = \frac{\pi}{3}$$

Now, let $\cot^{-1}(-\sqrt{3}) = y$

Hence,

$$\cot y = (-\sqrt{3})$$

$$= -\cot\left(\frac{\pi}{6}\right)$$

$$= \cot\left(\pi - \frac{\pi}{6}\right)$$

$$= \cot\left(\frac{5\pi}{6}\right)$$

Since, Range of principal value of $\cot^{-1} x = (0, \pi)$ Therefore,

$$\cot^{-1}(-\sqrt{3}) = \frac{5\pi}{6}$$

Then,

$$\begin{aligned} \tan^{-1} \sqrt{3} - \cot^{-1}(-\sqrt{3}) &= \frac{\pi}{3} - \frac{5\pi}{6} \\ &= -\frac{\pi}{2} \end{aligned}$$

Thus, the correct option is B.

Option (B) //

Q

1

Value of $\cos^{-1}(\cos x)$ outside principal range



$$\cos^{-1}\left(\cos \frac{13\pi}{6}\right)$$

Conclusion

$$\begin{aligned}\cos^{-1}\left(\cos \frac{13\pi}{6}\right) &= \cos^{-1}\left[\cos\left(2\pi + \frac{\pi}{6}\right)\right] \\ &= \cos^{-1}\left[\cos \frac{\pi}{6}\right] \\ &= \frac{\pi}{6}\end{aligned}$$

$$\frac{\pi}{6} //$$

Q

2

Value of $\tan^{-1}(\tan x)$ outside principal range



$$\tan^{-1}\left(\tan \frac{7\pi}{6}\right)$$

Conclusion

$$\begin{aligned}\tan^{-1}\left(\tan \frac{7\pi}{6}\right) &= \tan^{-1}\left[\tan\left(2\pi - \frac{5\pi}{6}\right)\right] \\ &= \tan^{-1}\left[-\tan\left(\frac{5\pi}{6}\right)\right] \\ &= \tan^{-1}\left[\tan\left(\pi - \frac{5\pi}{6}\right)\right] \\ &= \tan^{-1}\left[\tan \frac{\pi}{6}\right]\end{aligned}$$

$$= \frac{\pi}{6}$$

$$\frac{\pi}{6} //$$

Q

3

Prove sine-to-tangent double-angle identity



$$\text{Prove that } 2 \sin^{-1} \frac{3}{5} = \tan^{-1} \frac{24}{7}$$

Given

$$\text{Let } \sin^{-1} \frac{3}{5} = x \Rightarrow \sin x = \frac{3}{5}$$

Then,

$$\cos x = \sqrt{1 - \left(\frac{3}{5}\right)^2} = \frac{4}{5}$$

Therefore,

$$\tan x = \frac{3}{4}$$

$$x = \tan^{-1} \frac{3}{4}$$

$$\sin^{-1} \frac{3}{5} = \tan^{-1} \frac{3}{4}$$

Conclusion

Thus,

$$\begin{aligned}
 \text{LHS} &= 2 \sin^{-1} \frac{3}{5} \\
 &= 2 \tan^{-1} \frac{3}{4} \quad [\text{from (1)}] \\
 &= \tan^{-1} \left(\frac{2 \times \frac{3}{4}}{1 - \left(\frac{3}{4} \right)^2} \right) \\
 &= \tan^{-1} \left(\frac{24}{7} \right) \\
 &= \text{RHS}
 \end{aligned}$$

Q

4

Sum of two
inverse sines
as inverse
tangent

(())

Prove that $\sin^{-1} \frac{8}{17} + \sin^{-1} \frac{3}{5} = \tan^{-1} \frac{77}{36}$

Given

$$\text{Let } \sin^{-1} \frac{8}{17} = x \Rightarrow \sin x = \frac{8}{17}$$

Then,

$$\cos x = \sqrt{1 - \left(\frac{8}{17} \right)^2} = \sqrt{\left(\frac{225}{289} \right)} = \frac{15}{17}$$

Therefore,

$$\tan x = \frac{8}{15}$$

$$x = \tan^{-1} \frac{8}{15}$$

$$\sin^{-1} \frac{8}{17} = \tan^{-1} \frac{8}{15}$$

Substitute

$$\text{Now, let } \sin^{-1} \frac{3}{5} = y \Rightarrow \sin y = \frac{3}{5}$$

Then,

$$\cos y = \sqrt{1 - \left(\frac{3}{5} \right)^2} = \sqrt{\left(\frac{16}{25} \right)} = \frac{4}{5}$$

Therefore,

$$\tan y = \frac{3}{4}$$

$$y = \tan^{-1} \frac{3}{4}$$

$$\sin^{-1} \frac{3}{5} = \tan^{-1} \frac{3}{4}$$

Conclusion

Thus, by using (1) and (2)

$$\begin{aligned}
 \text{LHS} &= \sin^{-1} \frac{8}{17} + \sin^{-1} \frac{3}{5} \\
 &= \tan^{-1} \frac{8}{15} + \tan^{-1} \frac{3}{4} \\
 &= \tan^{-1} \left[\frac{\frac{8}{15} + \frac{3}{4}}{1 - \frac{8}{15} \cdot \frac{3}{4}} \right] \\
 &= \tan^{-1} \left[\frac{\frac{32+45}{60}}{\frac{60-24}{60}} \right] \\
 &= \tan^{-1} \frac{77}{36} \\
 &= \text{RHS}
 \end{aligned}$$

Q

5

Sum of two
inverse
cosines as
another
inverse
cosine



Prove that $\cos^{-1} \frac{4}{5} + \cos^{-1} \frac{12}{13} = \cos^{-1} \frac{33}{65}$

Given



$$\text{Let } \cos^{-1} \frac{4}{5} = x \Rightarrow \cos x = \frac{4}{5}$$

Then,

$$\sin x = \sqrt{1 - \left(\frac{4}{5}\right)^2} = \frac{3}{5}$$

Therefore,

$$\tan x = \frac{3}{4}$$

$$x = \tan^{-1} \frac{3}{4}$$

$$\cos^{-1} \frac{4}{5} = \tan^{-1} \frac{3}{4}$$

Substitute

$$\text{Now, let } \cos^{-1} \frac{12}{13} = y \Rightarrow \cos y = \frac{12}{13}$$

Then,

$$\sin y = \frac{5}{13}$$

Therefore,

$$\tan y = \frac{5}{12}$$

$$y = \tan^{-1} \frac{5}{12}$$

$$\cos^{-1} \frac{12}{13} = \tan^{-1} \frac{5}{12}$$

Thus, by using (1) and (2)

$$\begin{aligned}\cos^{-1} \frac{4}{5} + \cos^{-1} \frac{12}{13} &= \tan^{-1} \frac{3}{4} + \tan^{-1} \frac{5}{12} \\&= \tan^{-1} \left[\frac{\frac{3}{4} + \frac{5}{12}}{1 - \frac{3}{4} \cdot \frac{5}{12}} \right] \\&= \tan^{-1} \left[\frac{56}{33} \right]\end{aligned}$$

$$\text{Now, let } \cos^{-1} \frac{33}{65} = z \Rightarrow \cos z = \frac{33}{65}$$

Then,

$$\sin z = \sqrt{1 - \left(\frac{33}{65}\right)^2} = \frac{56}{65}$$

Therefore,

$$\begin{aligned}\tan z &= \frac{56}{33} \\z &= \tan^{-1} \frac{56}{33} \\ \cos^{-1} \frac{33}{65} &= \tan^{-1} \frac{56}{33}\end{aligned}$$

Conclusion

Thus, by using (3) and (4)

$$\cos^{-1} \frac{4}{5} + \cos^{-1} \frac{12}{13} = \cos^{-1} \frac{33}{65}$$

Hence proved.

Q

6

Sum of
inverse
cosine and
inverse sine
as another
inverse sine



Prove that $\cos^{-1} \frac{12}{13} + \sin^{-1} \frac{3}{5} = \sin^{-1} \frac{56}{65}$

Given



$$\text{Let } \cos^{-1} \frac{12}{13} = y \Rightarrow \cos y = \frac{12}{13}$$

Then,

$$\sin y = \sqrt{1 - \left(\frac{12}{13}\right)^2} = \frac{5}{13}$$

Therefore,

$$\begin{aligned}\tan y &= \frac{5}{12} \\y &= \tan^{-1} \frac{5}{12} \\ \cos^{-1} \frac{12}{13} &= \tan^{-1} \frac{5}{12}\end{aligned}$$

Substitute

$$\text{Now, let } \sin^{-1} \frac{3}{5} = x \Rightarrow \sin x = \frac{3}{5}$$

Then,

$$\cos x = \sqrt{1 - \left(\frac{3}{5}\right)^2} = \frac{4}{5}$$

Therefore,

$$\tan x = \frac{3}{4}$$

$$x = \tan^{-1} \frac{3}{4}$$

$$\sin^{-1} \frac{3}{5} = \tan^{-1} \frac{3}{4}$$

$$\text{Now, let } \sin^{-1} \frac{56}{65} = z \Rightarrow \sin z = \frac{56}{65}$$

Then,

$$\cos z = \sqrt{1 - \left(\frac{56}{65}\right)^2} = \frac{33}{65}$$

Therefore,

$$\tan z = \frac{56}{33}$$

$$z = \tan^{-1} \frac{56}{33}$$

$$\sin^{-1} \frac{56}{65} = \tan^{-1} \frac{56}{33}$$

Conclusion

Thus, by using (1) and (2)

$$\text{LHS} = \cos^{-1} \frac{12}{13} + \sin^{-1} \frac{3}{5}$$

$$= \tan^{-1} \frac{5}{12} + \tan^{-1} \frac{3}{4}$$

$$= \tan^{-1} \left[\frac{\frac{5}{12} + \frac{3}{4}}{1 - \frac{5}{12} \cdot \frac{3}{4}} \right]$$

$$= \tan^{-1} \left[\frac{\frac{5}{12} + \frac{3}{4}}{1 - \frac{5}{12} \cdot \frac{3}{4}} \right]$$

$$= \tan^{-1} \left[\frac{\frac{5}{12} + \frac{3}{4}}{1 - \frac{5}{12} \cdot \frac{3}{4}} \right]$$

$$= \tan^{-1} \left[\frac{\frac{5}{12} + \frac{3}{4}}{1 - \frac{5}{12} \cdot \frac{3}{4}} \right]$$

$$= \tan^{-1} \left(\frac{56}{33} \right)$$

$$= \sin^{-1} \frac{56}{65} \quad [\text{Using (3)}]$$

$$= \text{RHS}$$

Q

7

Inverse tangent as sum of inverse sine and inverse cosine



Prove that $\tan^{-1} \frac{63}{16} = \sin^{-1} \frac{5}{13} + \cos^{-1} \frac{3}{5}$

Given

$$\text{Let } \sin^{-1} \frac{5}{13} = x \Rightarrow \sin x = \frac{5}{13}$$

Then,

$$\cos x = \sqrt{1 - \left(\frac{5}{13}\right)^2} = \frac{12}{13}$$

Therefore,

$$\tan x = \frac{5}{12}$$

$$x = \tan^{-1} \frac{5}{12}$$

$$\sin^{-1} \frac{5}{13} = \tan^{-1} \frac{5}{12}$$

Substitute

$$\text{Now, let } \cos^{-1} \frac{3}{5} = y \Rightarrow \cos y = \frac{3}{5}$$

Then,

$$\sin y = \sqrt{1 - \left(\frac{3}{5}\right)^2} = \frac{4}{5}$$

Therefore,

$$\tan y = \frac{4}{3}$$

$$y = \tan^{-1} \frac{4}{3}$$

$$\cos^{-1} \frac{3}{5} = \tan^{-1} \frac{4}{3}$$

Conclusion

Thus, by using (1) and (2)

$$\text{RHS} = \sin^{-1} \frac{5}{13} + \cos^{-1} \frac{3}{5}$$

$$= \tan^{-1} \frac{5}{12} + \tan^{-1} \frac{4}{3}$$

$$= \tan^{-1} \left(\frac{\frac{5}{12} + \frac{4}{3}}{1 - \frac{5}{12} \cdot \frac{4}{3}} \right)$$

$$= \tan^{-1} \left(\frac{63}{16} \right)$$

$$= \text{L H S}$$

Q

8

**Inverse
tangent of $\sqrt{x}$**

(())

Prove that $\tan^{-1} \sqrt{x} = \frac{1}{2} \cos^{-1} \frac{1-x}{1+x}$, $x \in [0,1]$

Conclusion

$$\text{Let } x = \tan^2 \theta$$

Then,

$$\sqrt{x} = \tan \theta$$

$$\theta = \tan^{-1} \sqrt{x}$$

Therefore,

$$\begin{aligned} \left(\frac{1-x}{1+x} \right) &= \frac{1-\tan^2 \theta}{1+\tan^2 \theta} \\ &= \cos 2\theta \end{aligned}$$

Thus,

$$\begin{aligned}
 \text{RHS} &= \frac{1}{2} \cos^{-1} \left(\frac{1-x}{1+x} \right) \\
 &= \frac{1}{2} \cos^{-1}(\cos 2\theta) \\
 &= \frac{1}{2} \times 2\theta \\
 &= \theta \\
 &= \tan^{-1} \sqrt{x} \\
 &= \text{LHS}
 \end{aligned}$$

Q

9

Inverse
cotangent of
a
trigonometri
c ratio



Prove that $\cot^{-1} \left(\frac{\sqrt{1+\sin x} + \sqrt{1-\sin x}}{\sqrt{1+\sin x} - \sqrt{1-\sin x}} \right) = \frac{x}{2}, x \in (0, \frac{\pi}{4})$

Conclusion

$$\begin{aligned}
 &\left(\frac{\sqrt{1+\sin x} + \sqrt{1-\sin x}}{\sqrt{1+\sin x} - \sqrt{1-\sin x}} \right) = \frac{(\sqrt{1+\sin x} + \sqrt{1-\sin x})^2}{(\sqrt{1+\sin x})^2 - (\sqrt{1-\sin x})^2} \quad (\text{by rationalizing}) \\
 &= \frac{(1+\sin x) + (1-\sin x) + 2\sqrt{(1+\sin x)(1-\sin x)}}{1+\sin x - 1+\sin x} \\
 &= \frac{2(1+\sqrt{(1-\sin^2 x)})}{2 \sin x} = \frac{1+\cos x}{\sin x} \\
 &= \frac{2 \cos^2 \frac{x}{2}}{2 \sin \frac{x}{2} \cos \frac{x}{2}} \\
 &= \cot \frac{x}{2}
 \end{aligned}$$

Thus,

$$\begin{aligned}
 \text{L H S} &= \cot^{-1} \left(\frac{\sqrt{1+\sin x} + \sqrt{1-\sin x}}{\sqrt{1+\sin x} - \sqrt{1-\sin x}} \right) \\
 &= \cot^{-1} \left(\cot \frac{x}{2} \right) \\
 &= \frac{x}{2} \\
 &= \text{R H S}
 \end{aligned}$$

Q

10

Inverse
tangent of a
ratio as a
shifted half-
inverse-
cosine



Prove that $\tan^{-1} \left(\frac{\sqrt{1+x} - \sqrt{1-x}}{\sqrt{1+x} + \sqrt{1-x}} \right) = \frac{\pi}{4} - \frac{1}{2} \cos^{-1} x, -\frac{1}{\sqrt{2}} \leq x \leq 1$ [Hint: Put $x = \cos 2\theta$]

Conclusion

$$\text{Let } x = \cos 2\theta \Rightarrow \theta = \frac{1}{2} \cos^{-1} x$$

Thus,

$$\begin{aligned}
\text{LHS} &= \tan^{-1} \left(\frac{\sqrt{(1+x)} - \sqrt{(1-x)}}{\sqrt{(1+x)} + \sqrt{(1-x)}} \right) \\
&= \tan^{-1} \left(\frac{\sqrt{(1+\cos 2\theta)} - \sqrt{(1-\cos 2\theta)}}{\sqrt{(1+\cos 2\theta)} + \sqrt{(1-\cos 2\theta)}} \right) \\
&= \tan^{-1} \left(\frac{\sqrt{(2\cos^2 \theta)} - \sqrt{(2\sin^2 \theta)}}{\sqrt{(2\cos^2 \theta)} + \sqrt{(2\sin^2 \theta)}} \right) \\
&= \tan^{-1} \left(\frac{\sqrt{(2)} \cos \theta - \sqrt{(2)} \sin \theta}{\sqrt{(2)} \cos \theta + \sqrt{(2)} \sin \theta} \right) \\
&= \tan^{-1} \left(\frac{\cos \theta - \sin \theta}{\cos \theta + \sin \theta} \right) \\
&= \tan^{-1} \left(\frac{1 - \tan \theta}{1 + \tan \theta} \right) \\
&= \tan^{-1} 1 - \tan^{-1}(\tan \theta) \\
&= \frac{\pi}{4} - \theta \\
&= \frac{\pi}{4} - \frac{1}{2} \cos^{-1} x \\
&= \text{RHS}
\end{aligned}$$

Q

11

Solve an
inverse trig
equation



Solve $2 \tan^{-1}(\cos x) = \tan^{-1}(2 \operatorname{cosec} x)$

Conclusion

It is given that $2 \tan^{-1}(\cos x) = \tan^{-1}(2 \operatorname{cosec} x)$

Since, $2 \tan^{-1}(x) = \tan^{-1} \frac{2x}{1-x^2}$

Hence,

$$\Rightarrow \tan^{-1} \left(\frac{2 \cos x}{1 - \cos^2 x} \right) = \tan^{-1}(2 \operatorname{cosec} x)$$

$$\Rightarrow \left(\frac{2 \cos x}{1 - \cos^2 x} \right) = (2 \operatorname{cosec} x)$$

$$\Rightarrow \frac{2 \cos x}{\sin^2 x} = \frac{2}{\sin x}$$

$$\Rightarrow \cos x = \sin x$$

$$\Rightarrow \tan x = 1$$

$$\Rightarrow \tan x = \tan \frac{\pi}{4}$$

Therefore,

$$x = n\pi + \frac{\pi}{4}, \text{ where } n \in \mathbb{Z}.$$

$$\boxed{x = n\pi + \frac{\pi}{4}, n \in \mathbb{Z}} //$$

Q

12

Solve an
inverse trig
equation



Solve $\tan^{-1} \frac{1-x}{1+x} = \frac{1}{2} \tan^{-1} x, (x > 0)$

Conclusion

Since $\tan^{-1} x - \tan^{-1} y = \tan^{-1} \frac{x-y}{1+xy}$

Hence,

$$\Rightarrow \tan^{-1} \frac{1-x}{1+x} = \frac{1}{2} \tan^{-1} x$$

$$\Rightarrow \tan^{-1} 1 - \tan^{-1} x = \frac{1}{2} \tan^{-1} x$$

$$\Rightarrow \frac{\pi}{4} = \frac{3}{2} \tan^{-1} x$$

$$\Rightarrow \tan^{-1} x = \frac{\pi}{6}$$

$$\Rightarrow x = \tan \frac{\pi}{6}$$

$$\Rightarrow x = \frac{1}{\sqrt{3}}$$

$$x = \frac{1}{\sqrt{3}} //$$

Q

13

Value of
 $\sin(\tan^{-1} x)$
(multiple
choice)



$\sin(\tan^{-1} x), |x| < 1$ is equal to (A) $\frac{x}{\sqrt{1-x^2}}$ (B) $\frac{1}{\sqrt{1-x^2}}$ (C) $\frac{1}{\sqrt{1+x^2}}$

(D) $\frac{x}{\sqrt{1+x^2}}$

Conclusion

Let $\tan y = x$

Therefore,

$$\sin y = \frac{x}{\sqrt{1+x^2}}$$

Now, let $\tan^{-1} x = y$

Therefore,

$$y = \sin^{-1} \left(\frac{x}{\sqrt{1+x^2}} \right)$$

Hence,

$$\tan^{-1} x = \sin^{-1} \left(\frac{x}{\sqrt{1+x^2}} \right)$$

Thus,

$$\begin{aligned} \sin(\tan^{-1} x) &= \sin \left(\sin^{-1} \left(\frac{x}{\sqrt{1+x^2}} \right) \right) \\ &= \frac{x}{\sqrt{1+x^2}} \end{aligned}$$

Thus, the correct option is D.

Option (D) //

Q

14

Solve an
inverse-sine
equation
(multiple
choice)



Solve: $\sin^{-1}(1-x) - 2 \sin^{-1} x = \frac{\pi}{2}$, then x is equal to (A) 0, $\frac{1}{2}$ (B) 1, $\frac{1}{2}$ (C) 0

(D) $\frac{1}{2}$

Conclusion

It is given that $\sin^{-1}(1-x) - 2 \sin^{-1} x = \frac{\pi}{2}$

$$\Rightarrow \sin^{-1}(1-x) - 2 \sin^{-1} x = \frac{\pi}{2}$$

$$\Rightarrow -2 \sin^{-1} x = \frac{\pi}{2} - \sin^{-1}(1-x)$$

$$\Rightarrow -2 \sin^{-1} x = \cos^{-1}(1-x)$$

Let $\sin^{-1} x = y \Rightarrow \sin y = x$

Hence,

$$\cos y = \sqrt{1-x^2}$$

$$y = \cos^{-1}(\sqrt{1-x^2})$$

$$\sin^{-1} x = \cos^{-1} \sqrt{1-x^2}$$

From equation (1), we have

$$-2 \cos^{-1} \sqrt{1-x^2} = \cos^{-1}(1-x)$$

Put $x = \sin y$

$$\Rightarrow -2 \cos^{-1} \sqrt{1-\sin^2 y} = \cos^{-1}(1-\sin y)$$

$$\Rightarrow -2 \cos^{-1}(\cos y) = \cos^{-1}(1-\sin y)$$

$$\Rightarrow -2 y = \cos^{-1}(1-\sin y)$$

$$\Rightarrow 1-\sin y = \cos(-2 y)$$

$$\Rightarrow 1-\sin y = \cos 2 y$$

$$\Rightarrow 1-\sin y = 1-2 \sin^2 y$$

$$\Rightarrow 2 \sin^2 y - \sin y = 0$$

$$\Rightarrow \sin y(2 \sin y - 1) = 0$$

$$\Rightarrow \sin y = 0, \frac{1}{2}$$

Therefore,

$$x = 0, \frac{1}{2}$$

When $x = \frac{1}{2}$, it does not satisfy the equation.

Hence, $x = 0$ is the only solution

Thus, the correct option is C.

Option (C) ✓✓